

`remove_proc_entry()` in the destructor, `proc_win_exit()`, in chronological reverse order.

For every entry created under `/proc`, a corresponding struct `proc_dir_entry` is created. For each, many of its fields could be further updated as needed:

- `mode`—Permissions of the file
- `uid`—User ID of the file
- `gid`—Group ID of the file

Additionally, for a regular file, the following two function pointers for reading and writing over the file could be provided, respectively:

- `int (*read_proc)(char *page, char **start, off_t off, int count, int *eof, void *data)`
- `int (*write_proc)(struct file *file, const char __user *buffer, unsigned long count, void *data)`

`write_proc()` is very similar to the character driver's file operation `write()`. The above implementation lets the user write a digit from 0 to 9, and accordingly sets the internal state. `read_proc()` in the above implementation provides the current state, and the time since the system has been booted up—in different units, based on the current state. These are jiffies in state 0; milliseconds in state 1; seconds and milliseconds in state 2; hours, minutes and seconds in state 3; and <not implemented> in other states. And to check the computation accuracy, Figure 2 highlights the system uptime in the output of `top`. `read_proc`'s `page` parameter is a page-sized buffer, typically to be filled up with `count` bytes from offset `off`. But more often than not (because of less content), just the

`page` is filled up, ignoring all other parameters.

All the `/proc`-related structure definitions and function declarations are available through `<linux/proc_fs.h>`. The jiffies-related function declarations and macro definitions are in `<linux/jiffies.h>`. As a special note, the actual jiffies are calculated by subtracting `INITIAL_JIFFIES`, since on boot-up, jiffies is initialised to `INITIAL_JIFFIES` instead of zero.

Summing up

"Hey Pugs! Why did you set the folder name to *anil*? Who is this Anil? You could have used my name, or maybe yours," suggested Shweta. "Ha! That's a surprise. My real name is Anil; it's just that everyone in college knows me as Pugs," smiled Pugs.

Watch out for further technical romancing from Pugs a.k.a. Anil. 

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